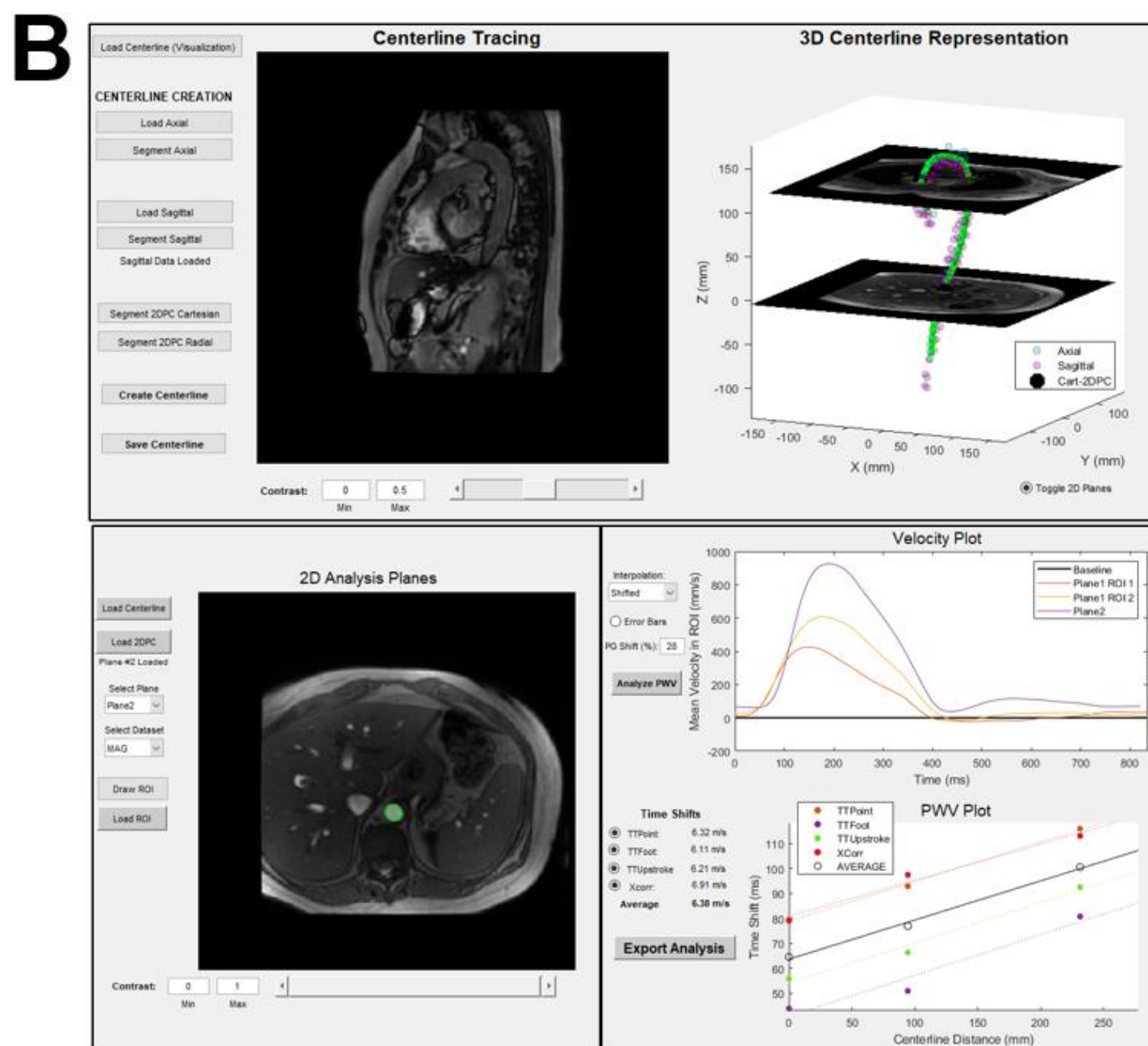
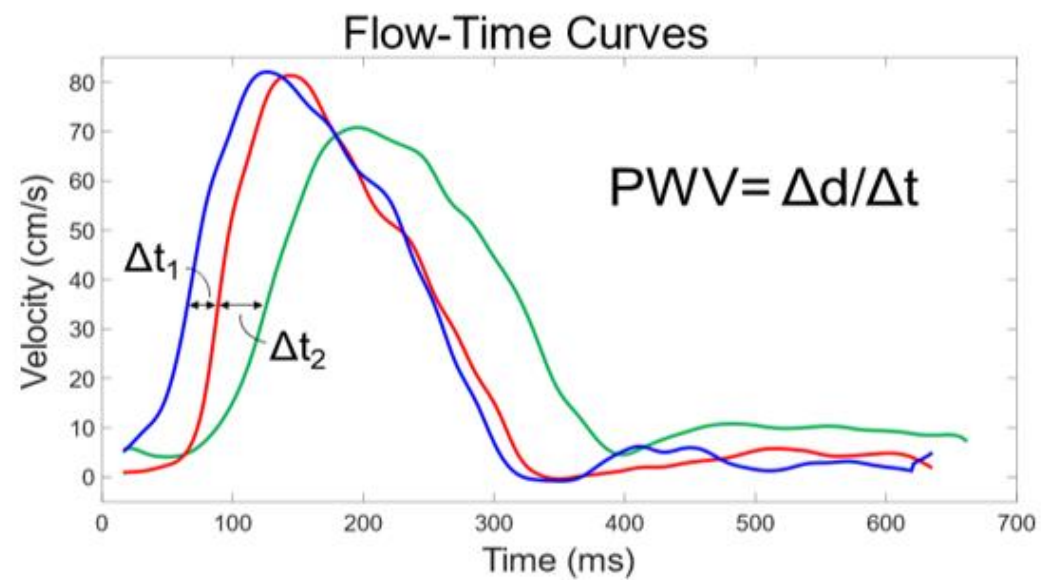
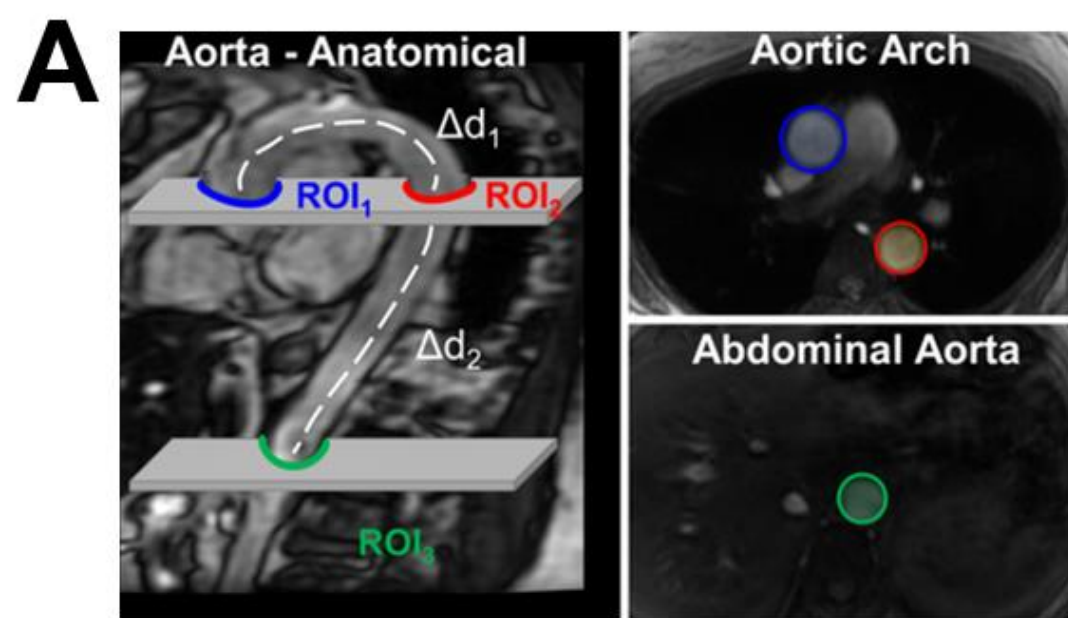
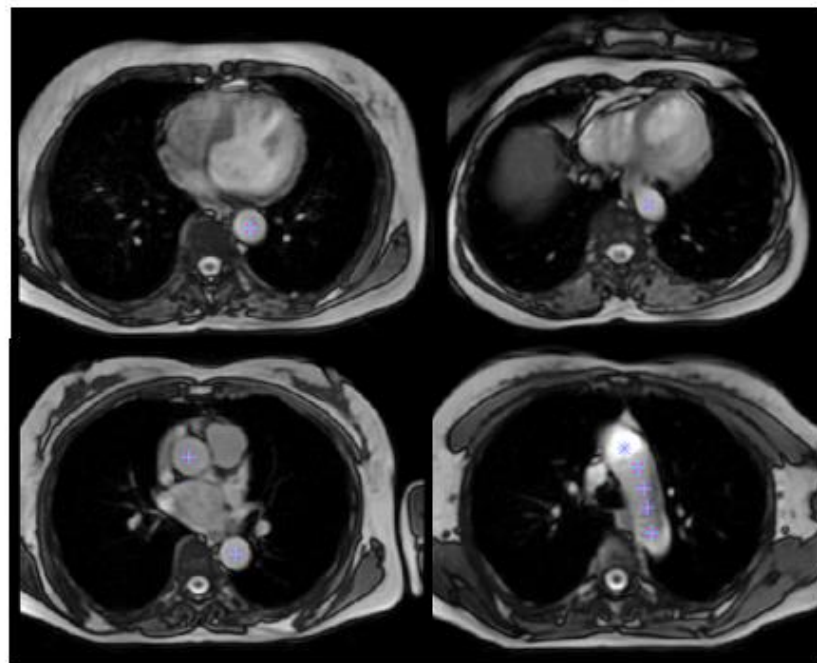
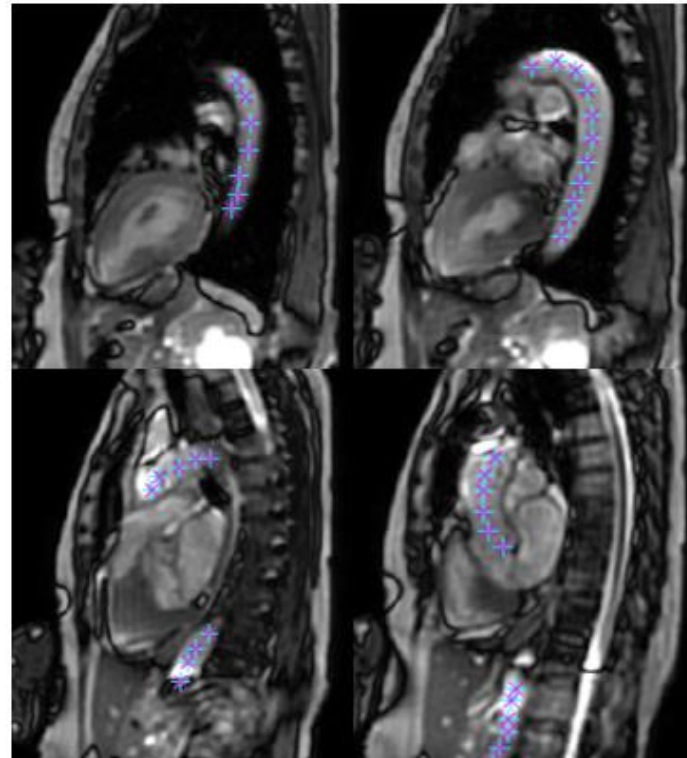




Axial bSSFP – Aorta Point Selection



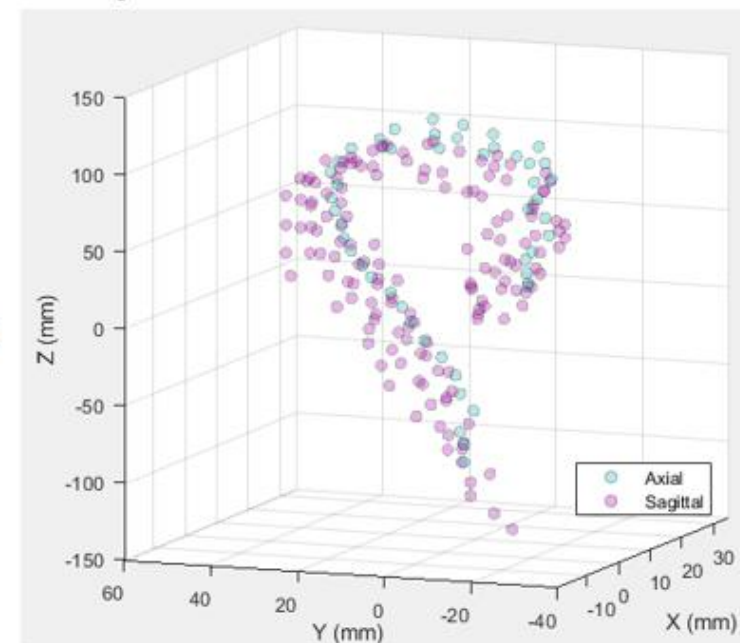
Sagittal bSSFP – Aorta Point Selection



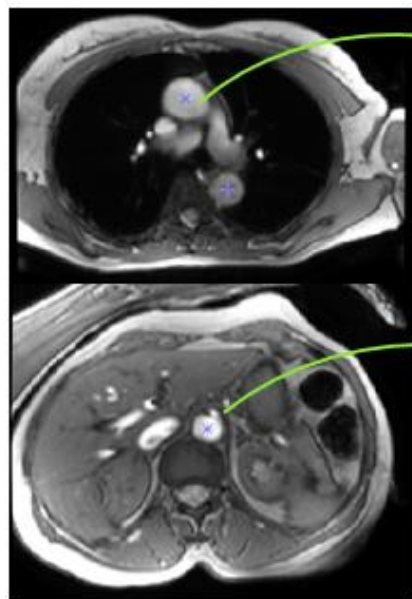
+



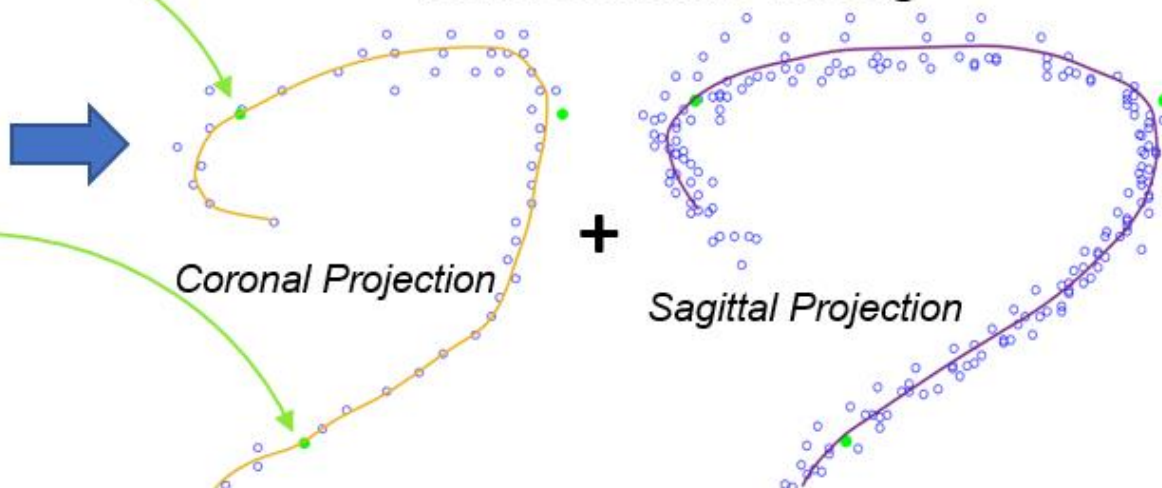
Spatial Localization of Points



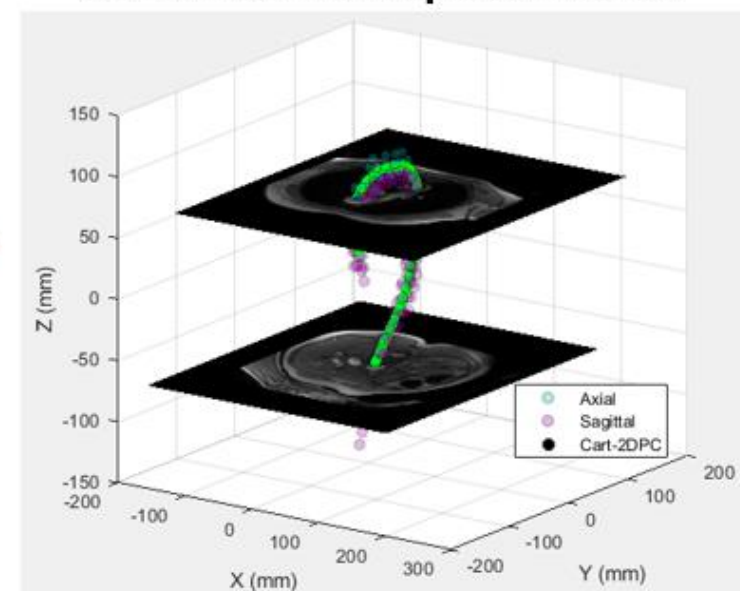
2DPC Point Selection



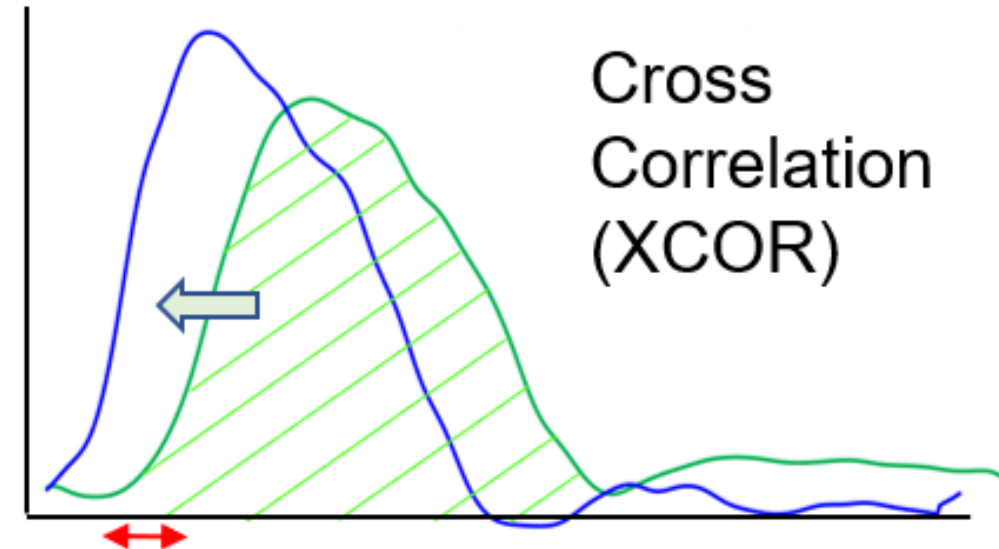
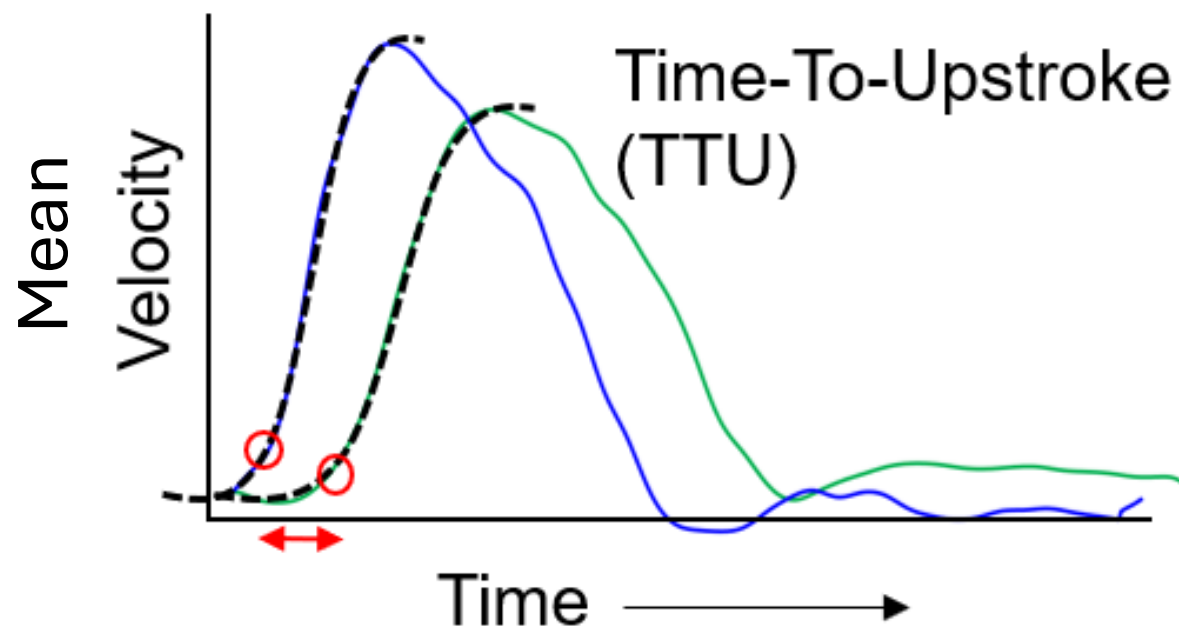
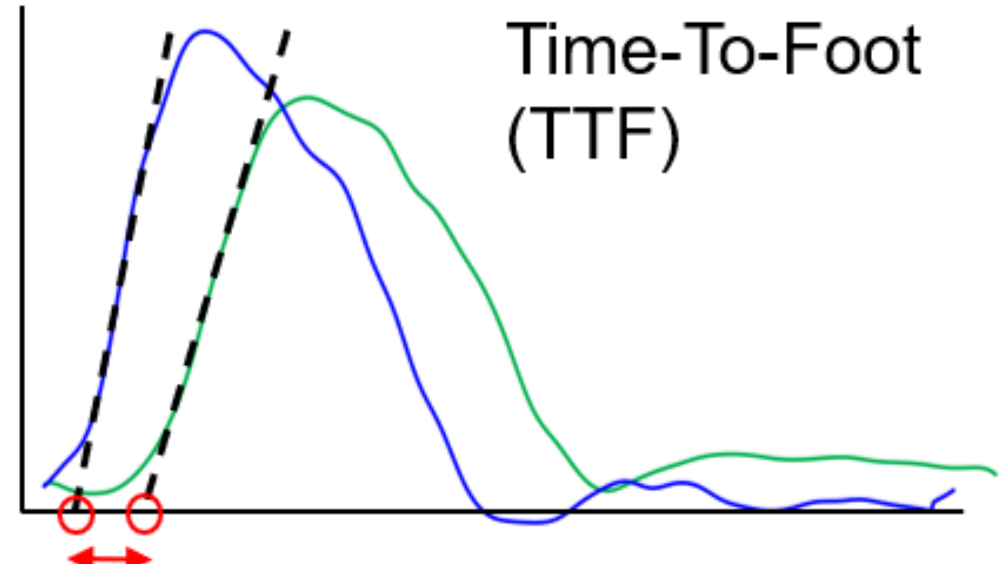
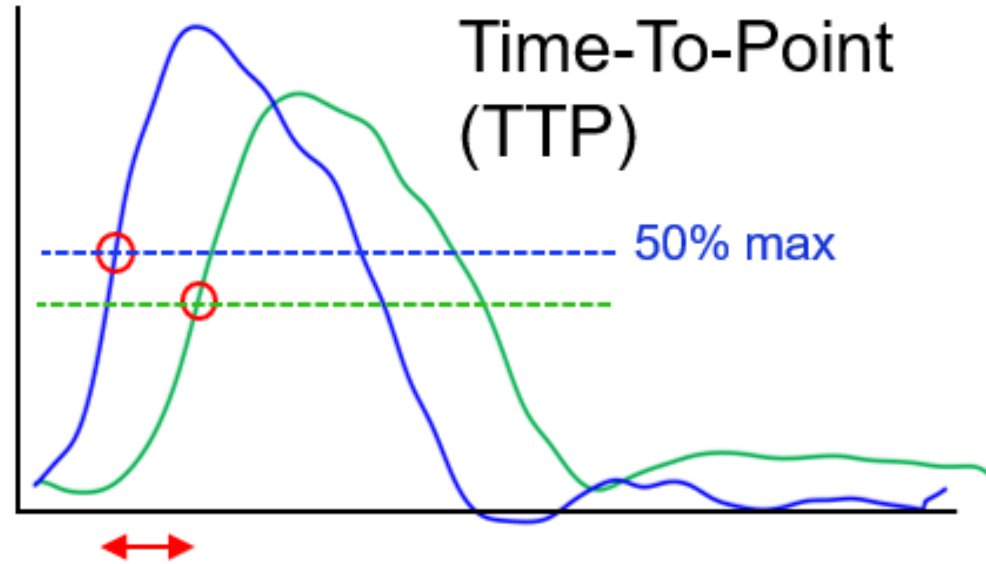
Aorta Centerline Tracing

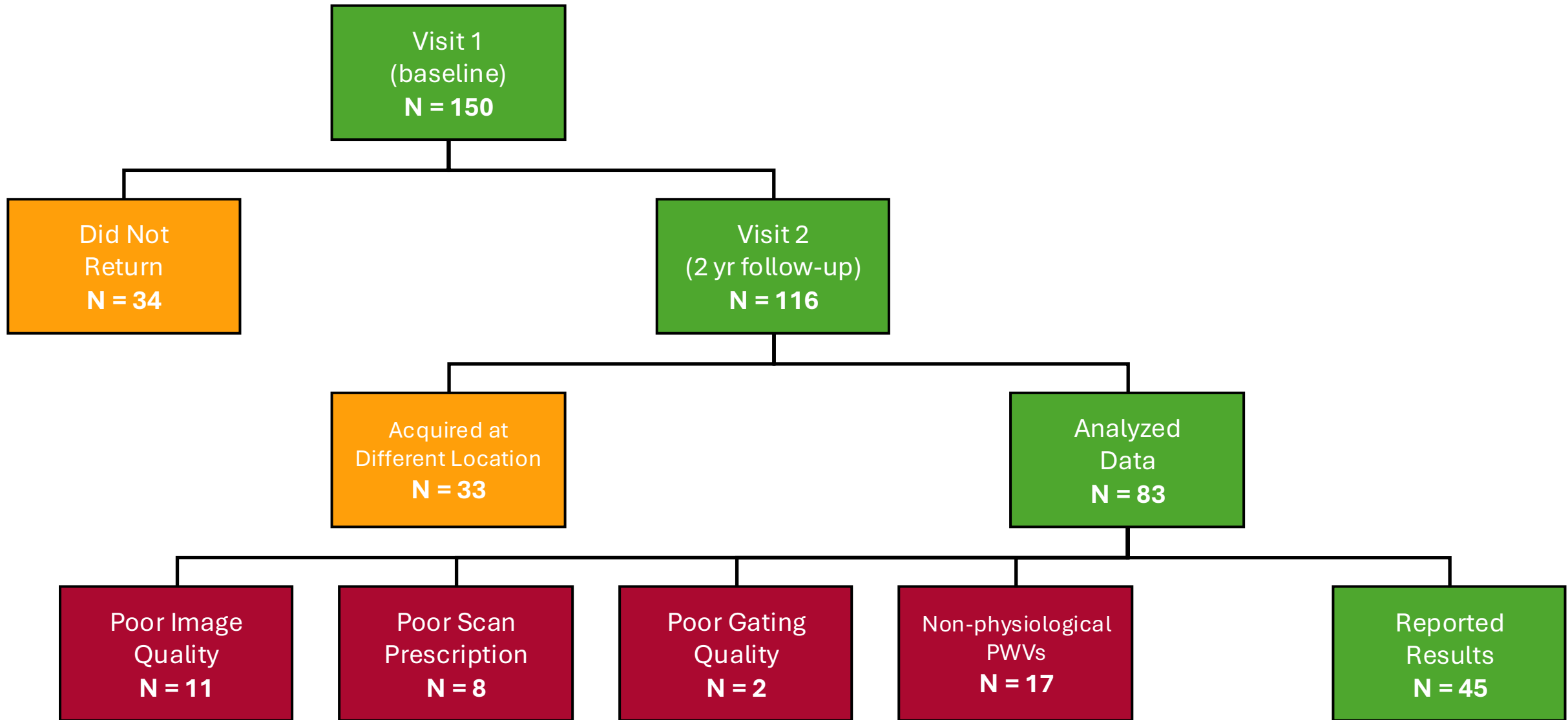


3D Centerline Representation



Transit Time (Time Shift) Methods





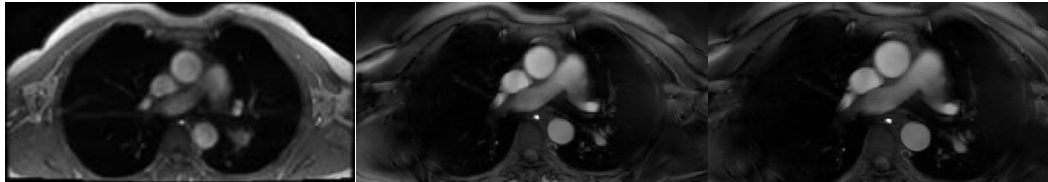
Time-Averaged

Cartesian

Stand. Radial

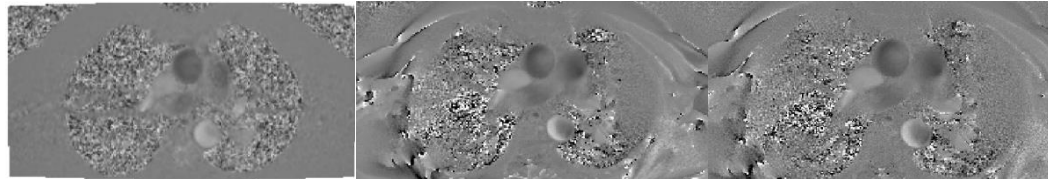
Hi-Res Radial

Magnitude



AAo

Phase

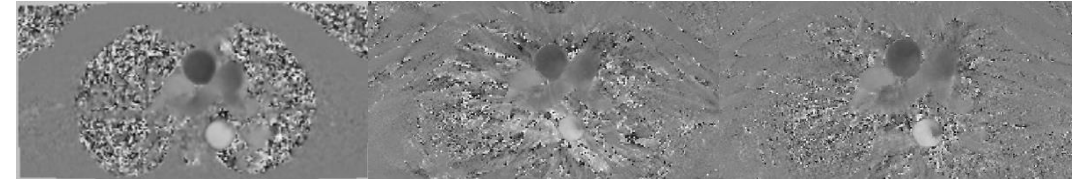
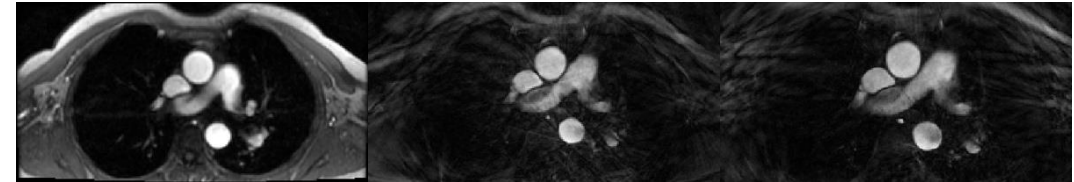


Single Frame (Peak Systole)

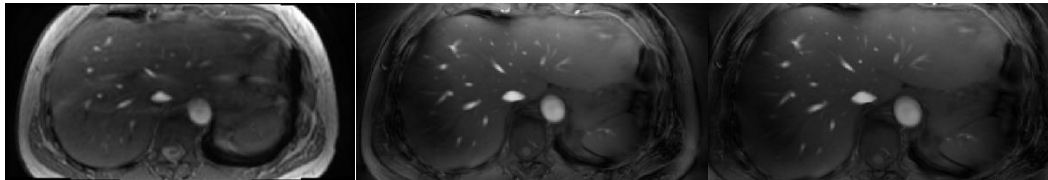
Cartesian

Stand. Radial

Hi-Res Radial

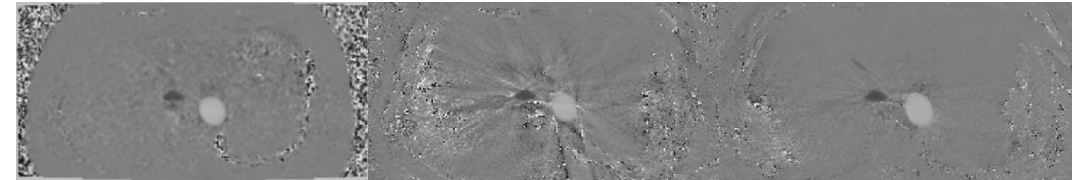
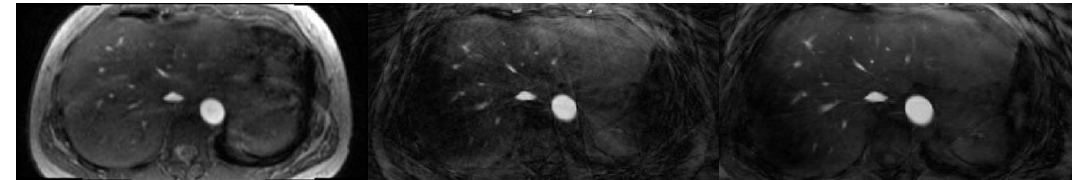
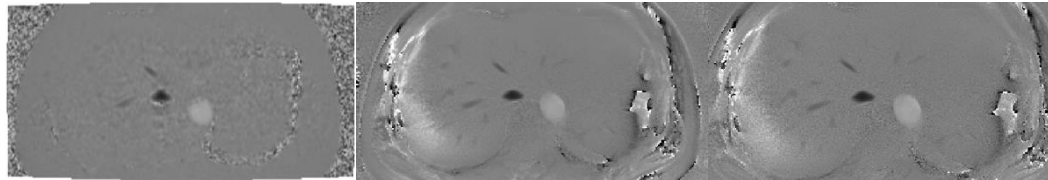


Magnitude



AbdAo

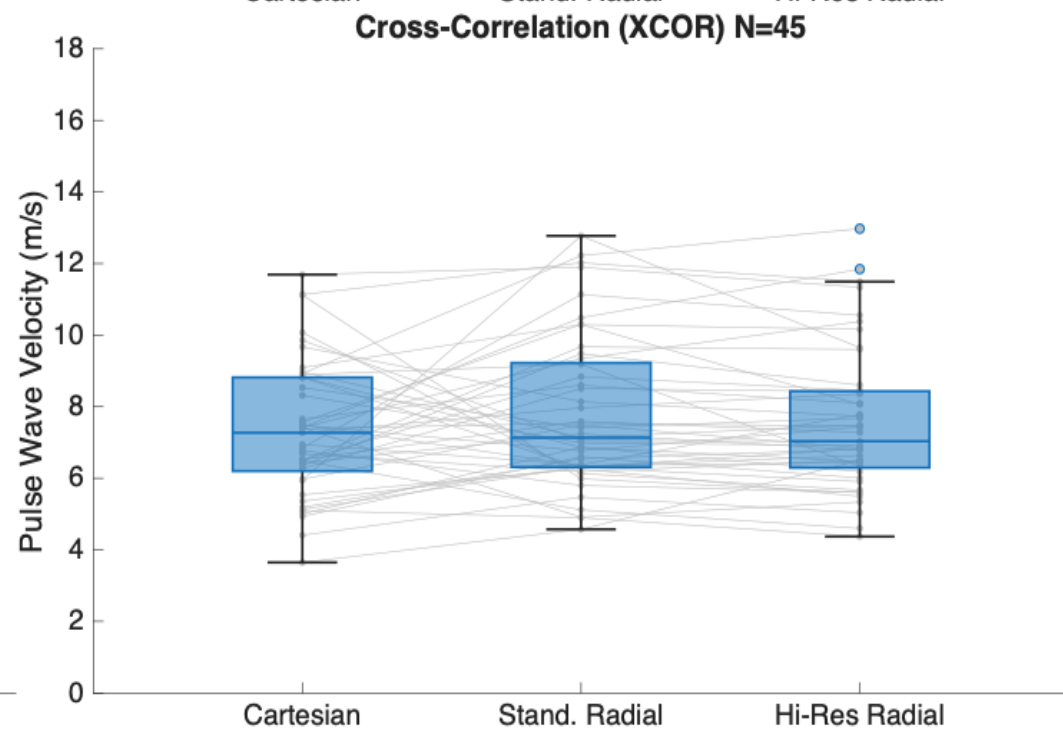
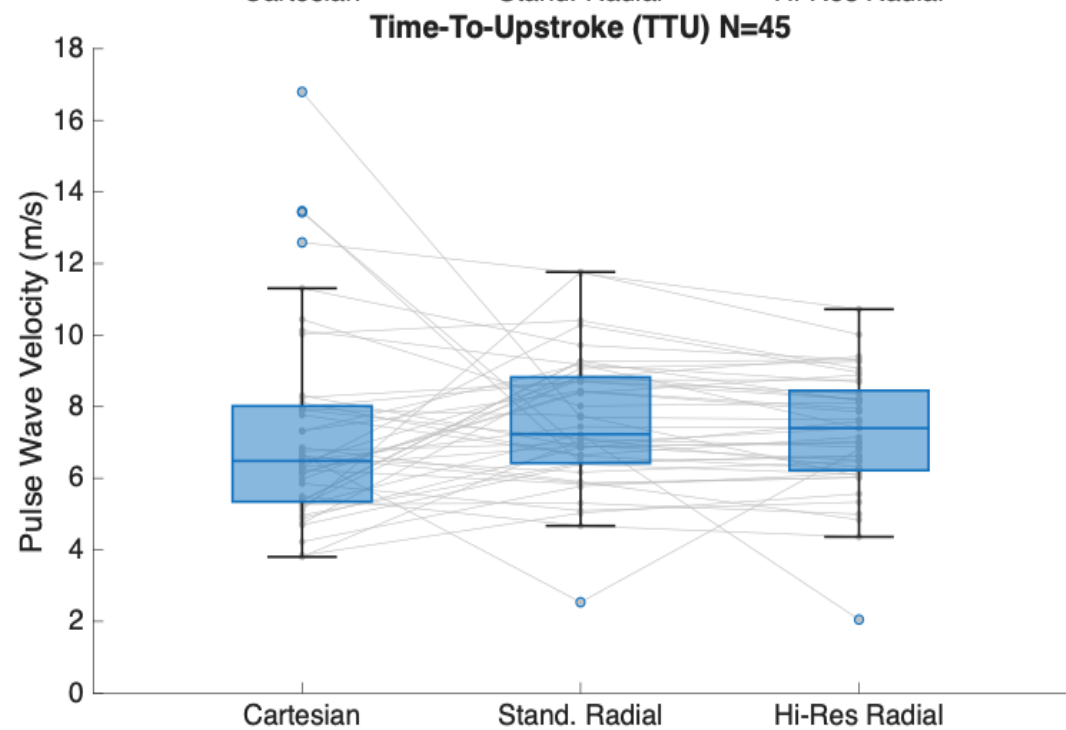
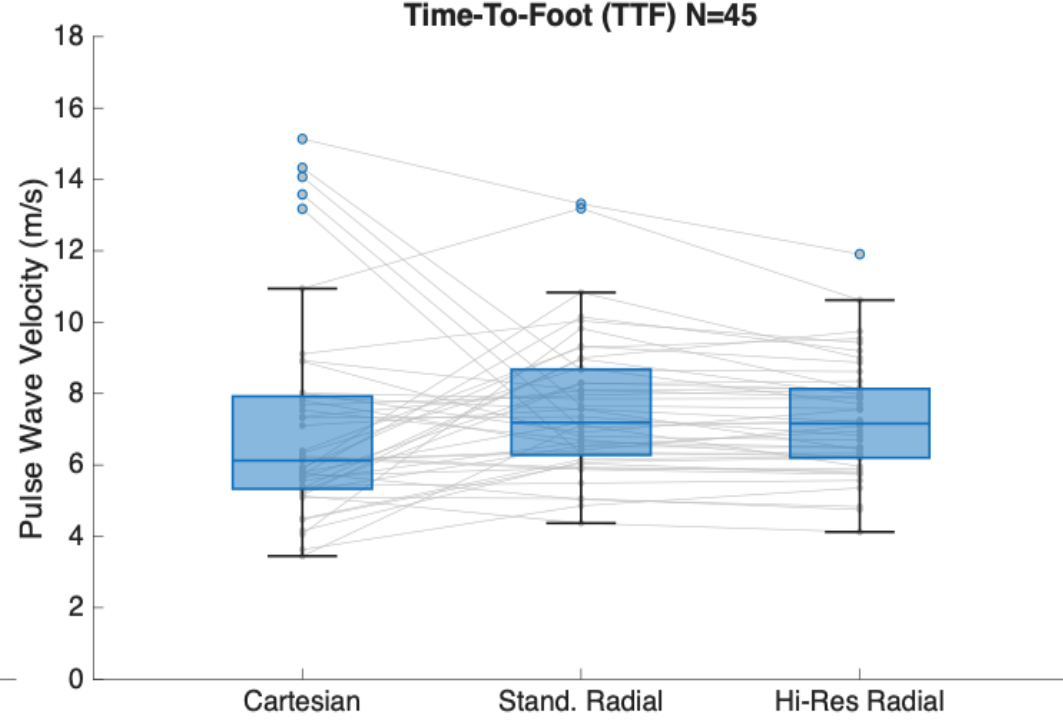
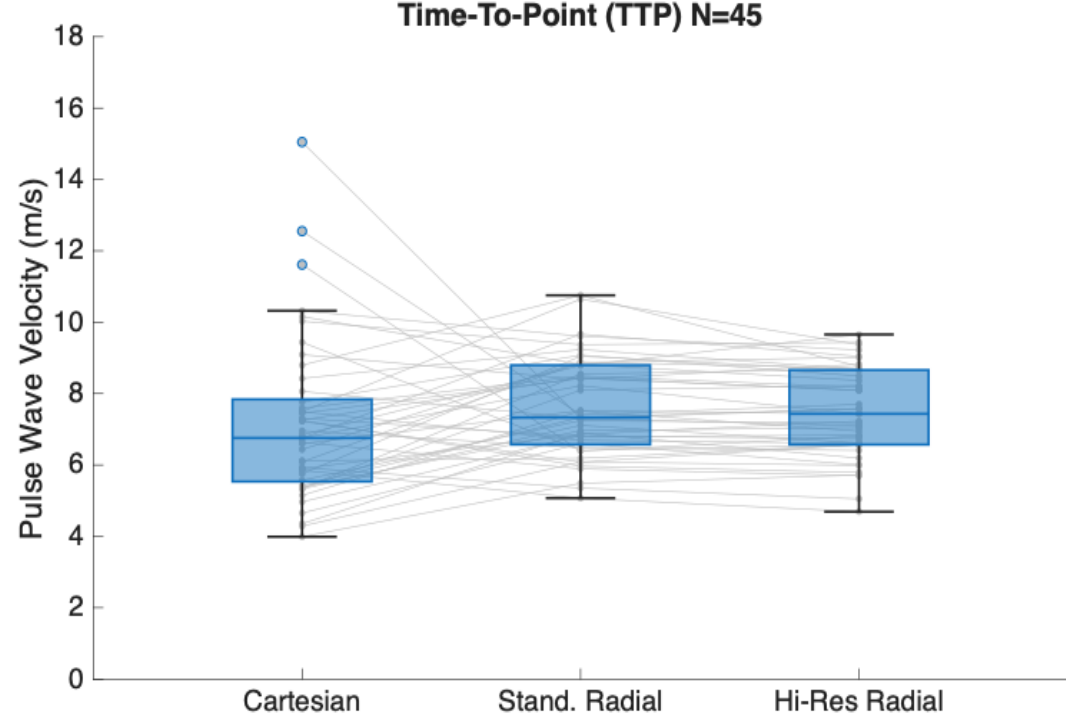
Phase

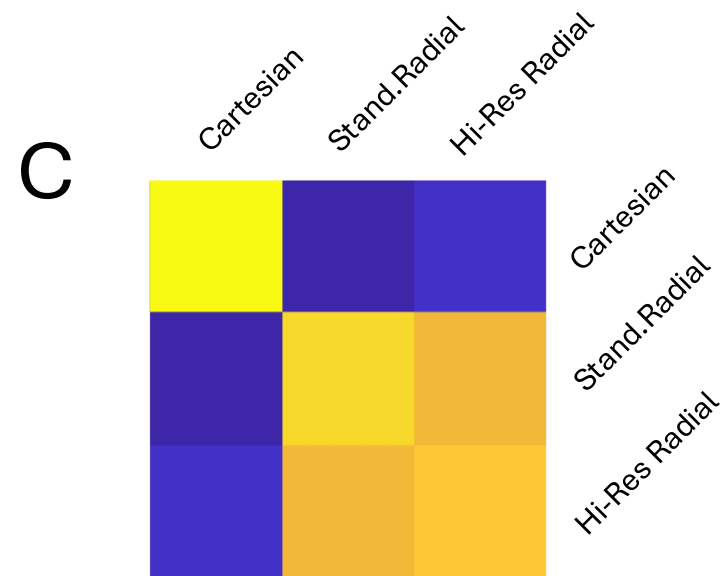
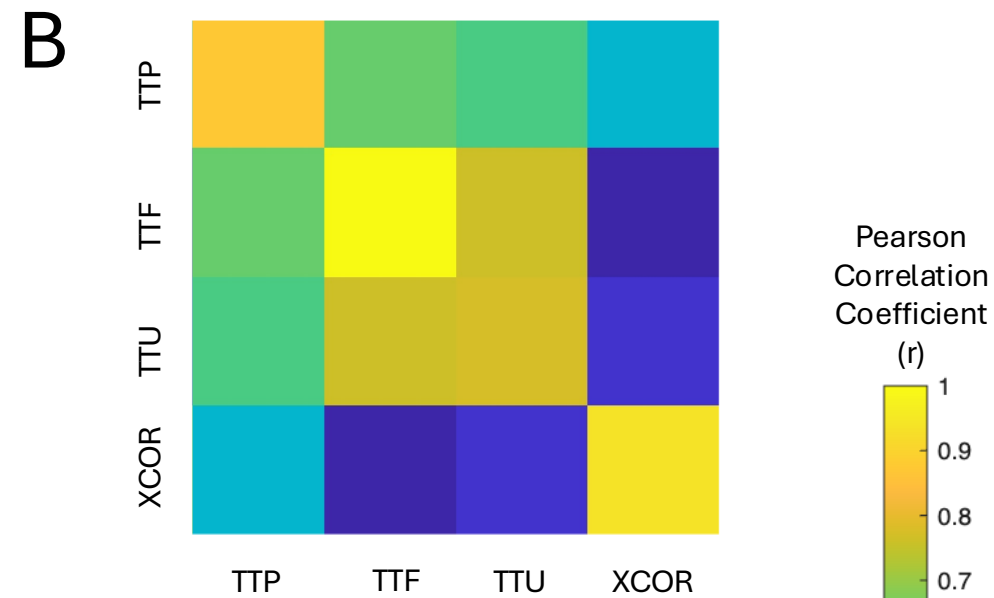
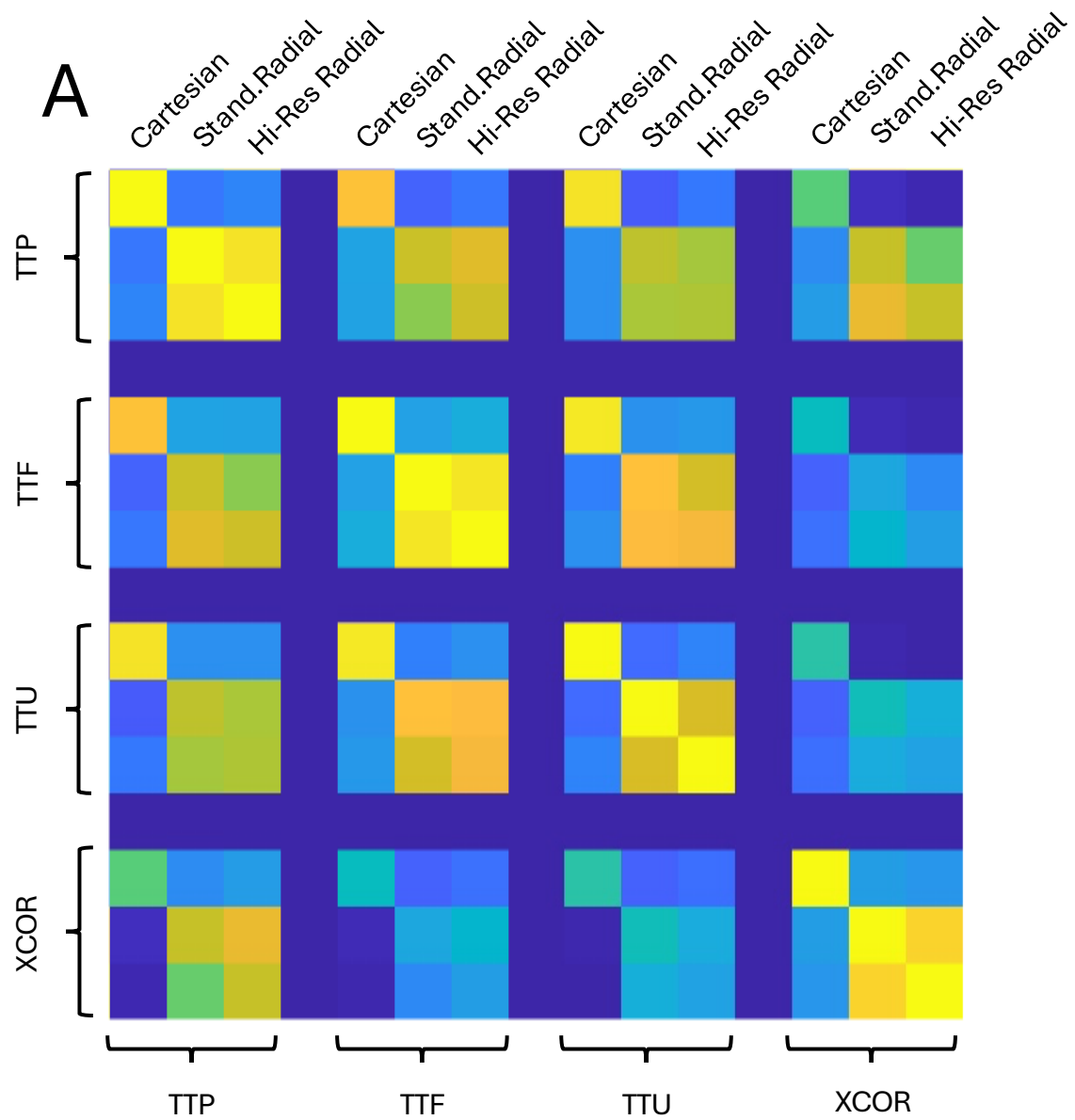


-150 cm/s

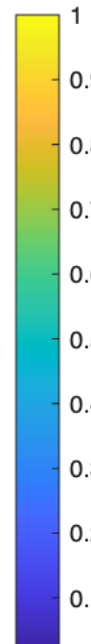


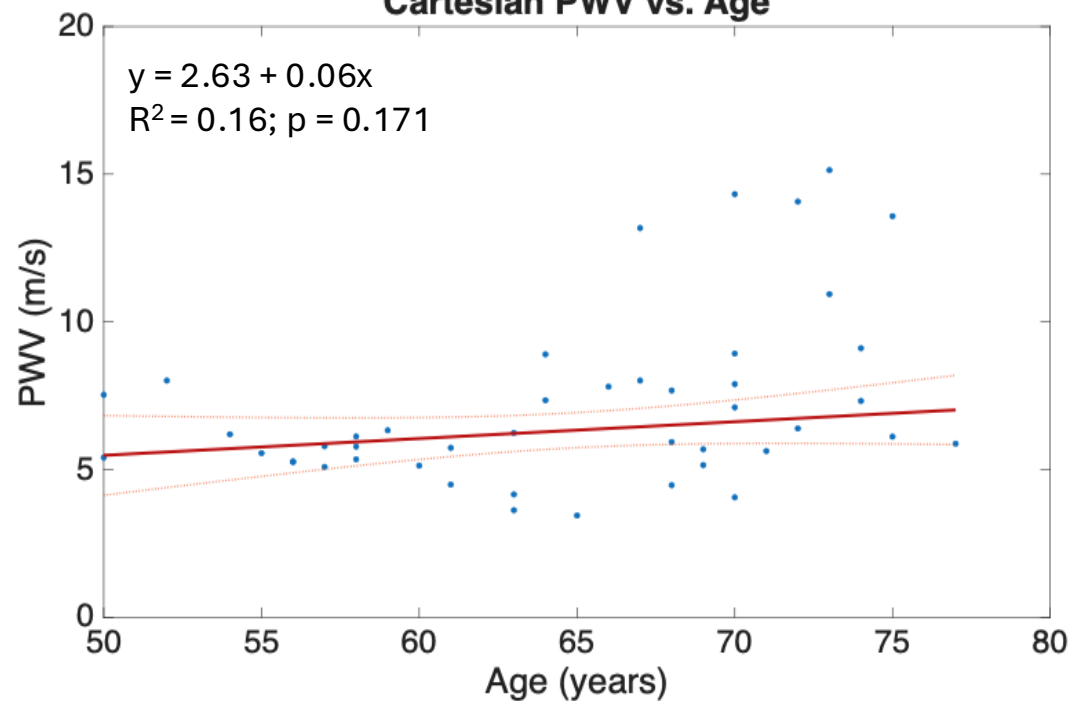
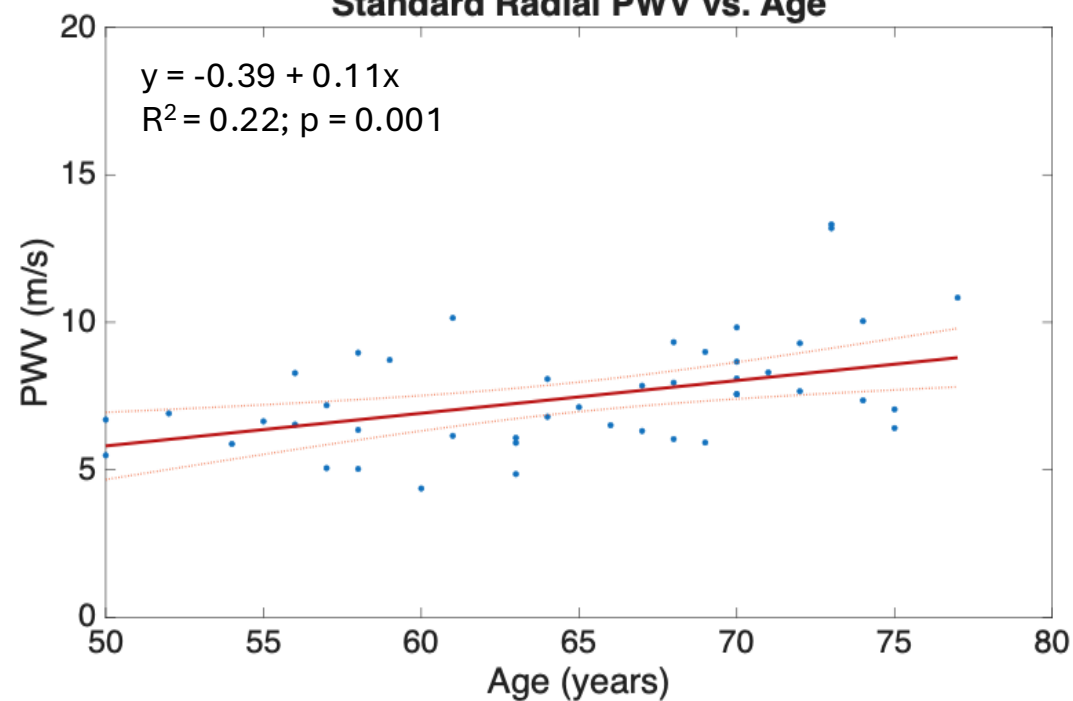
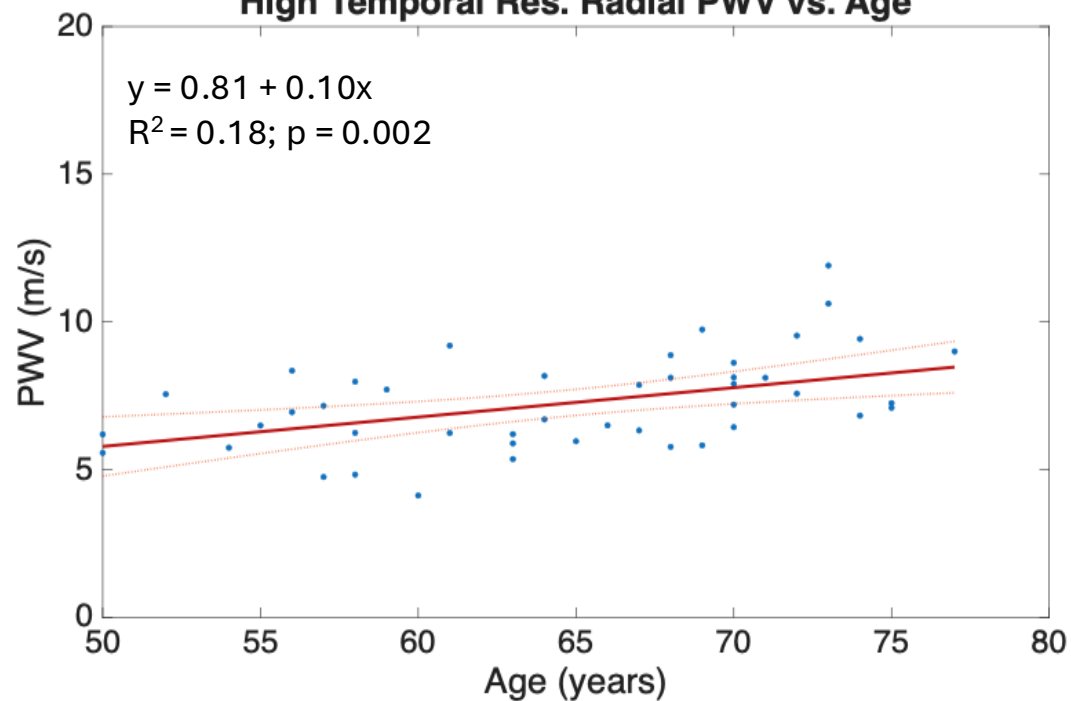
+150 cm/s



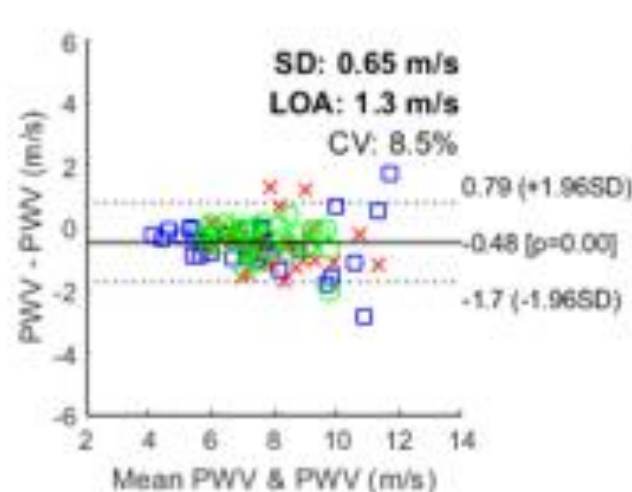
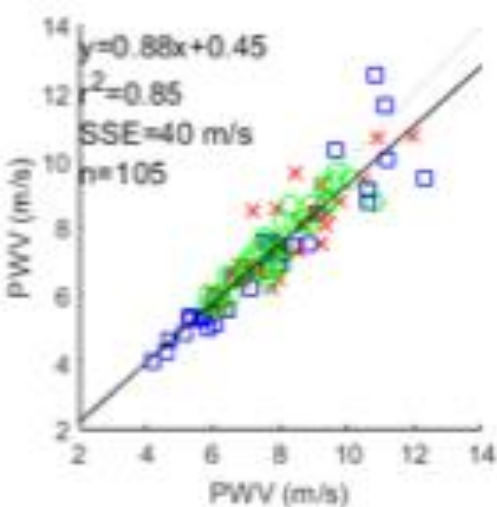


Pearson
Correlation
Coefficient
(r)

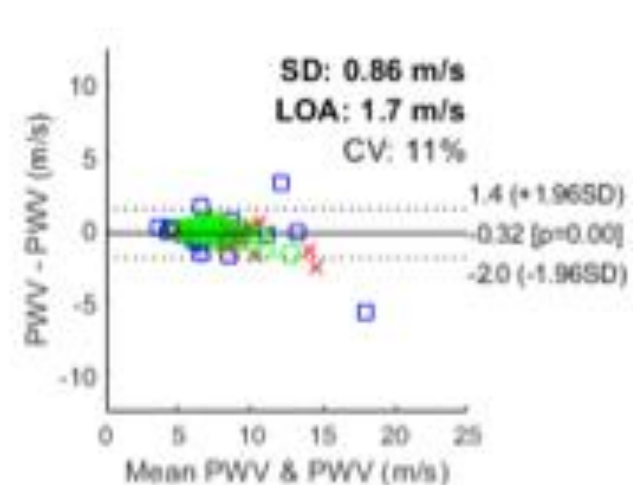
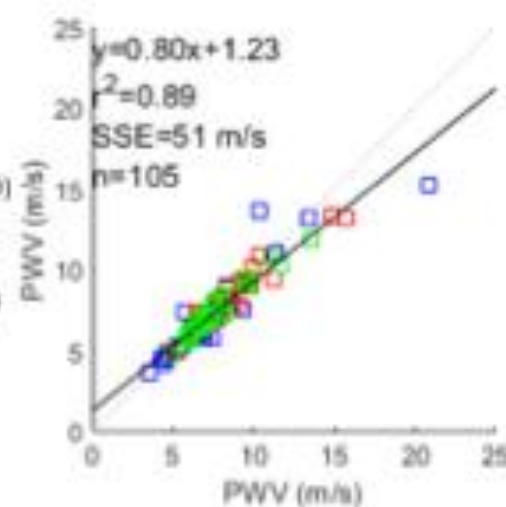


Cartesian PWV vs. Age**Standard Radial PWV vs. Age****High Temporal Res. Radial PWV vs. Age**

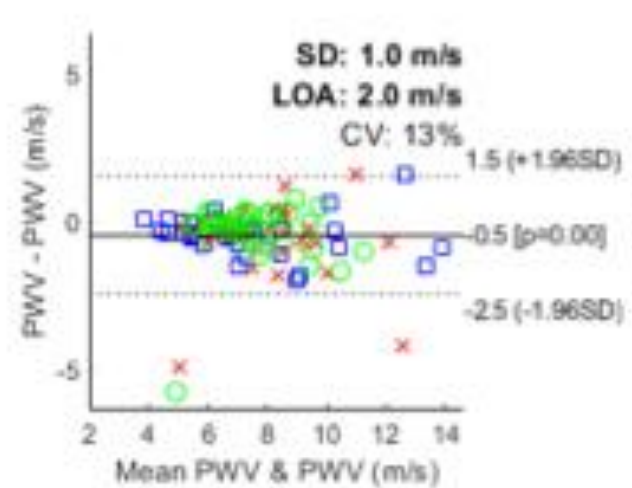
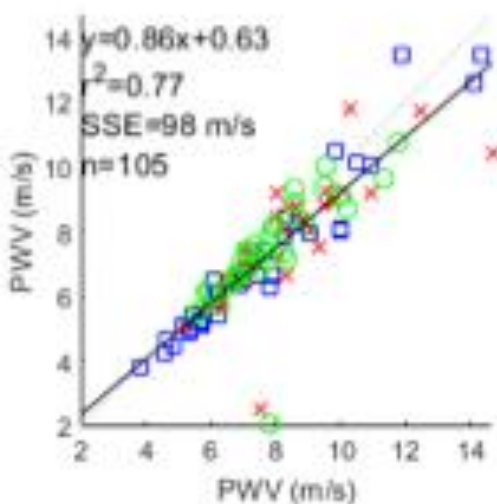
Time-To-Point (TTP) N=35



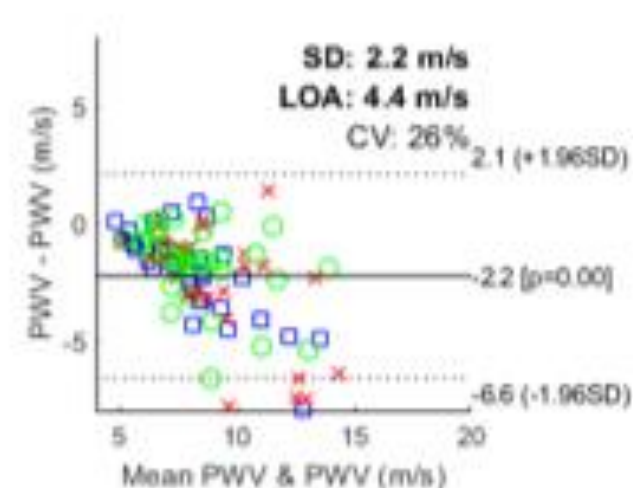
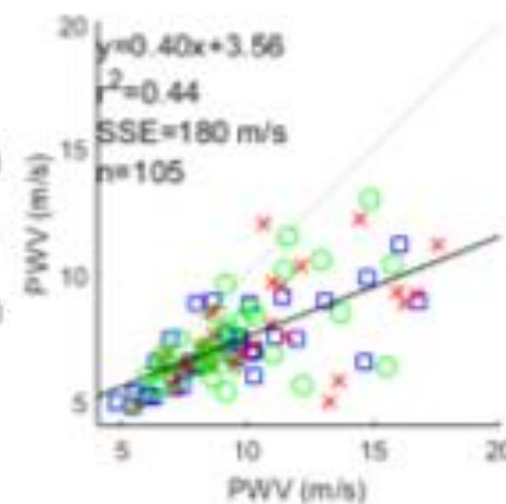
Time-To-Foot (TTF) N=35



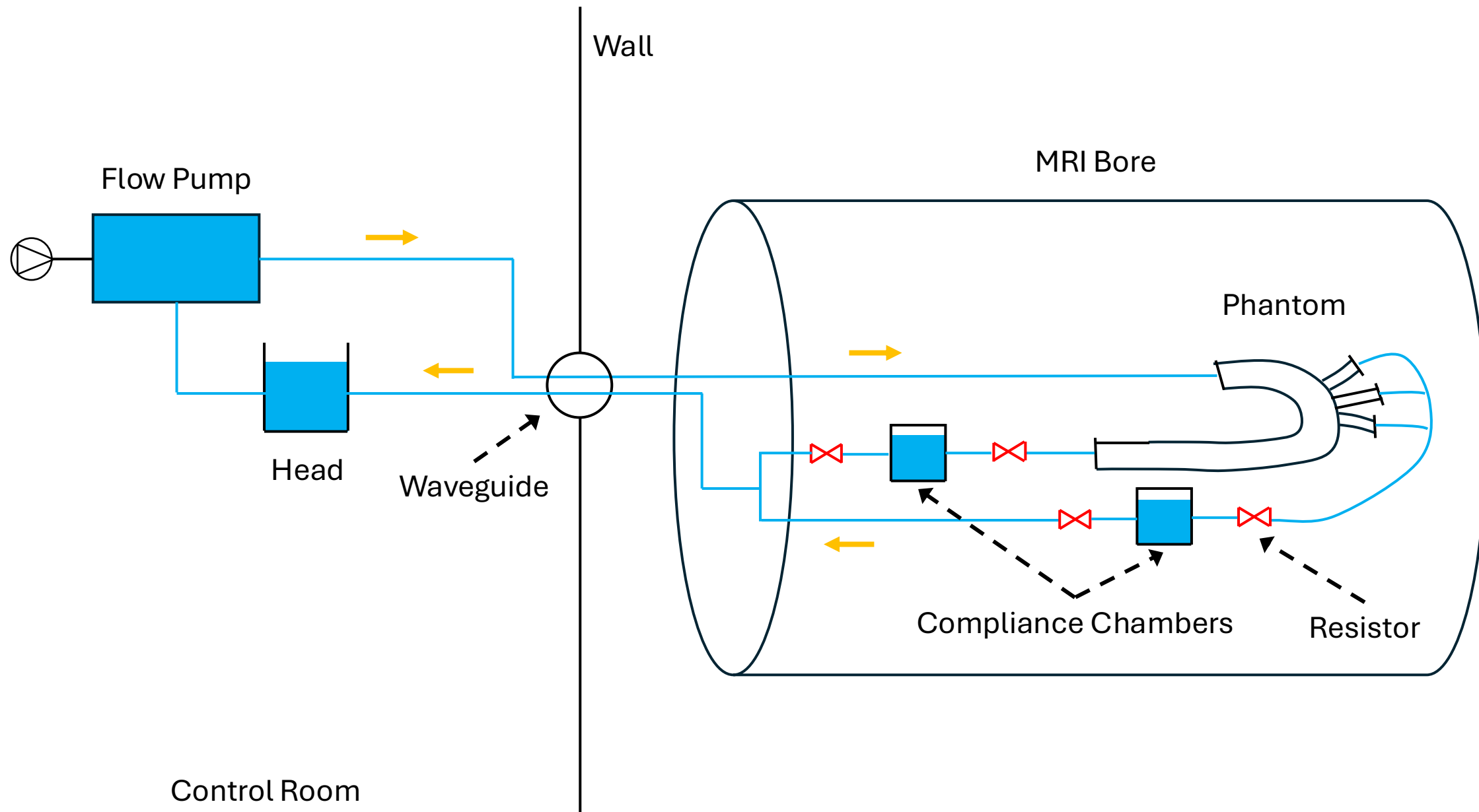
Time-To-Upstroke (TTU) N=35



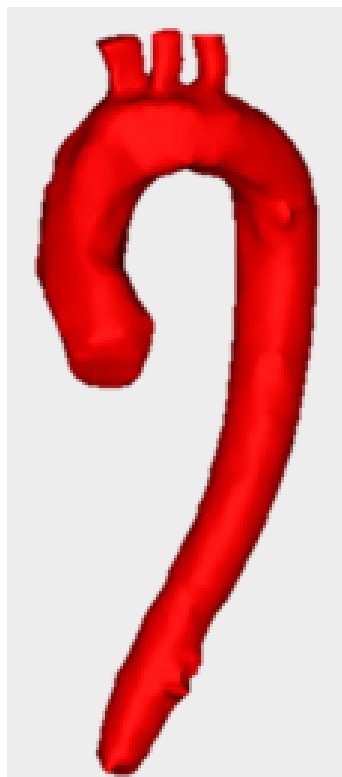
Cross Correlation (XCOR) N=35



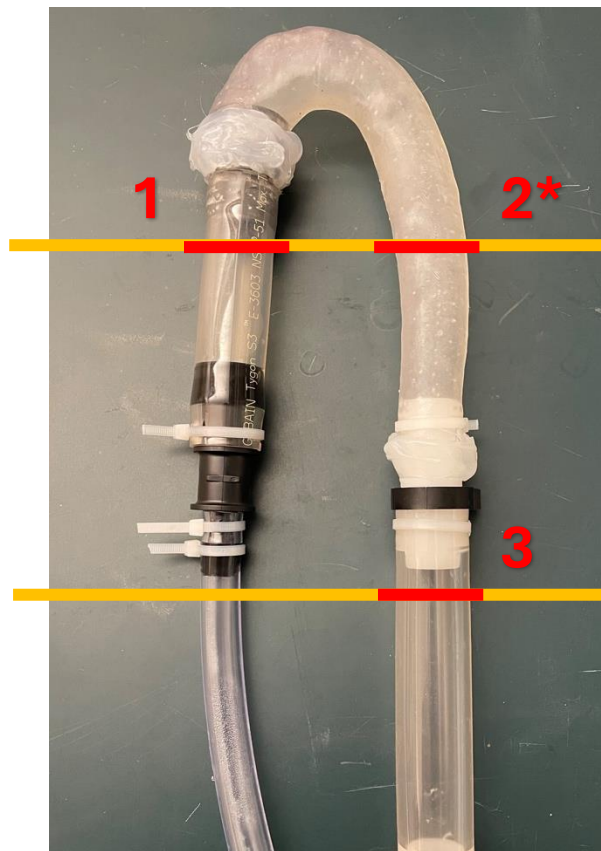
□ Cartesian
 × Radial Low-Res
 ○ Radial High-Res



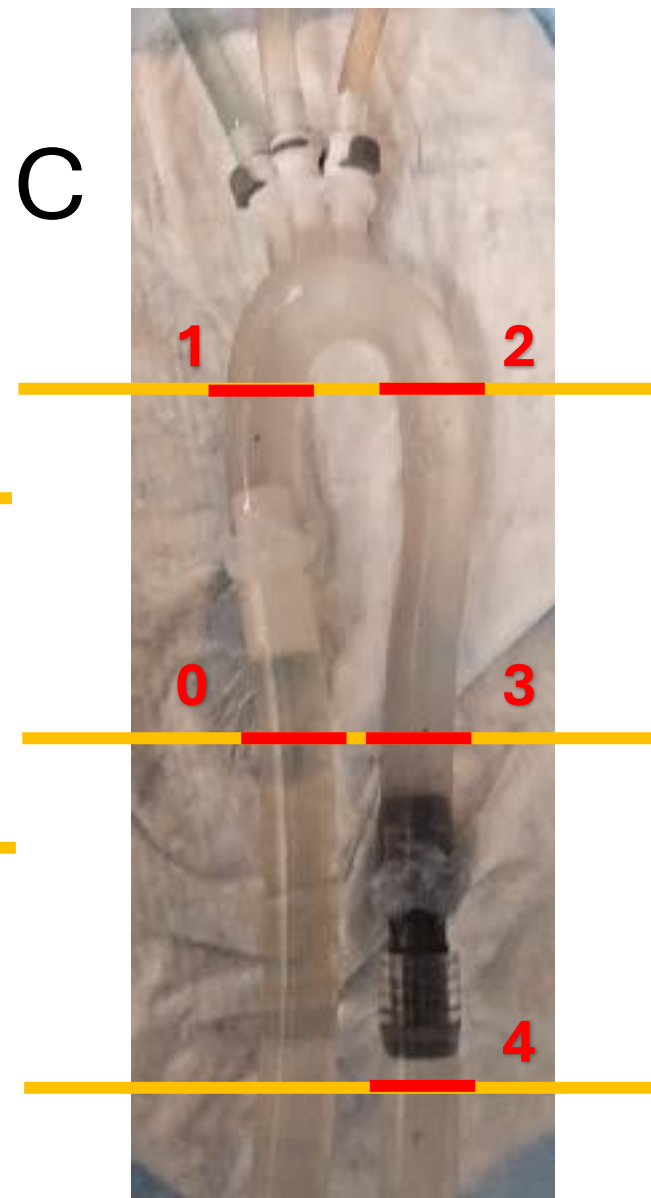
A



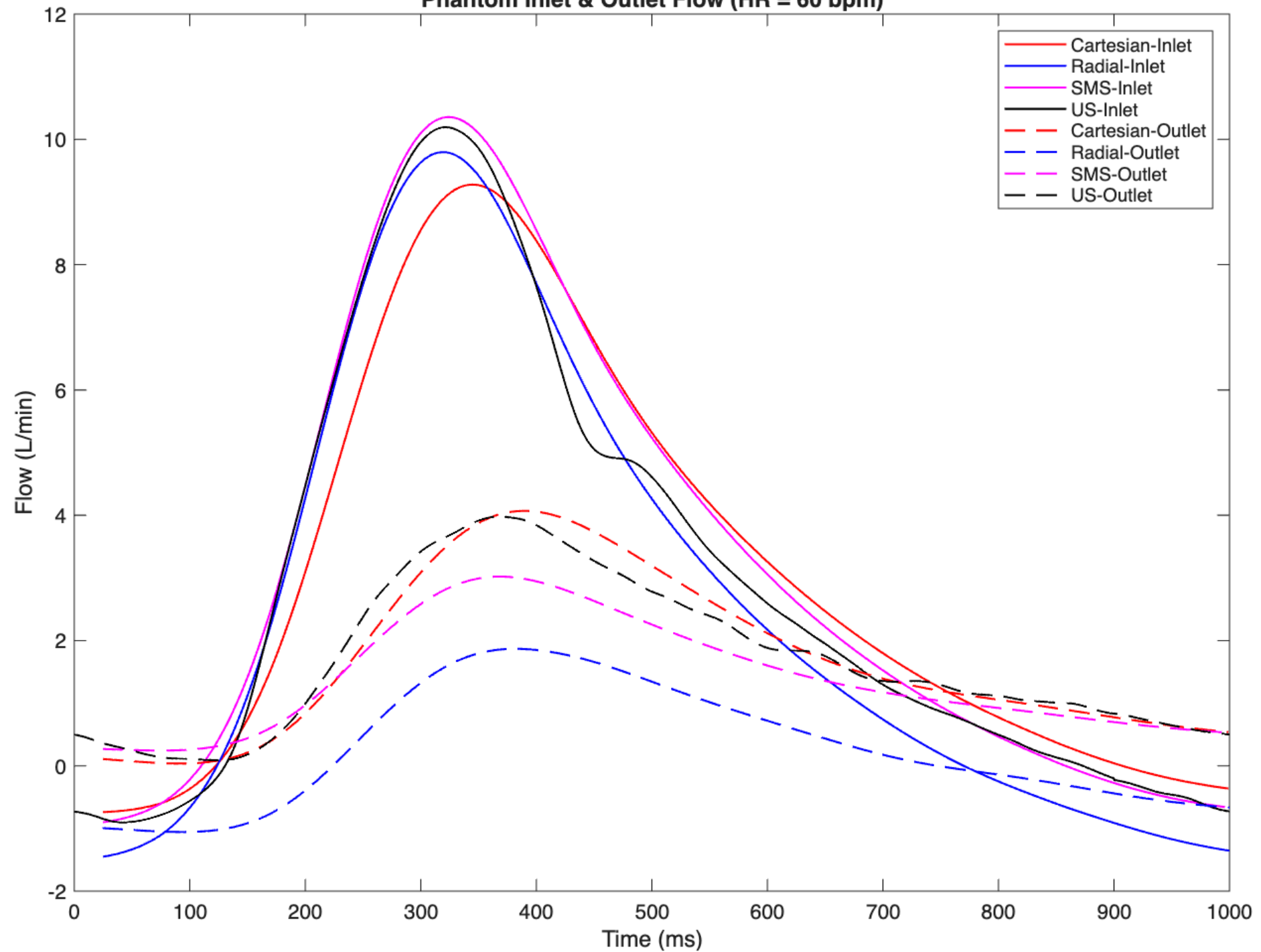
B



C



Phantom Inlet & Outlet Flow (HR = 60 bpm)

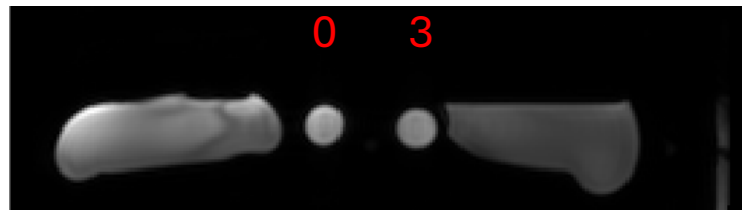
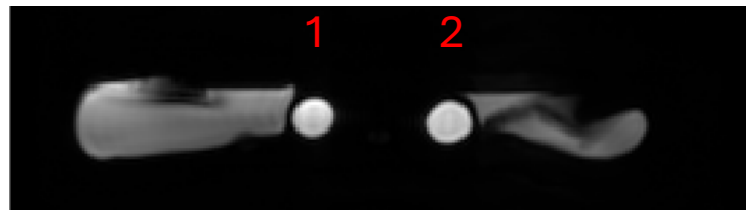


Aortic Arch

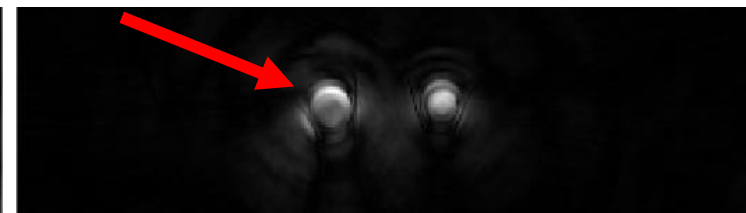
Abdominal Aorta

Outlet

Cartesian

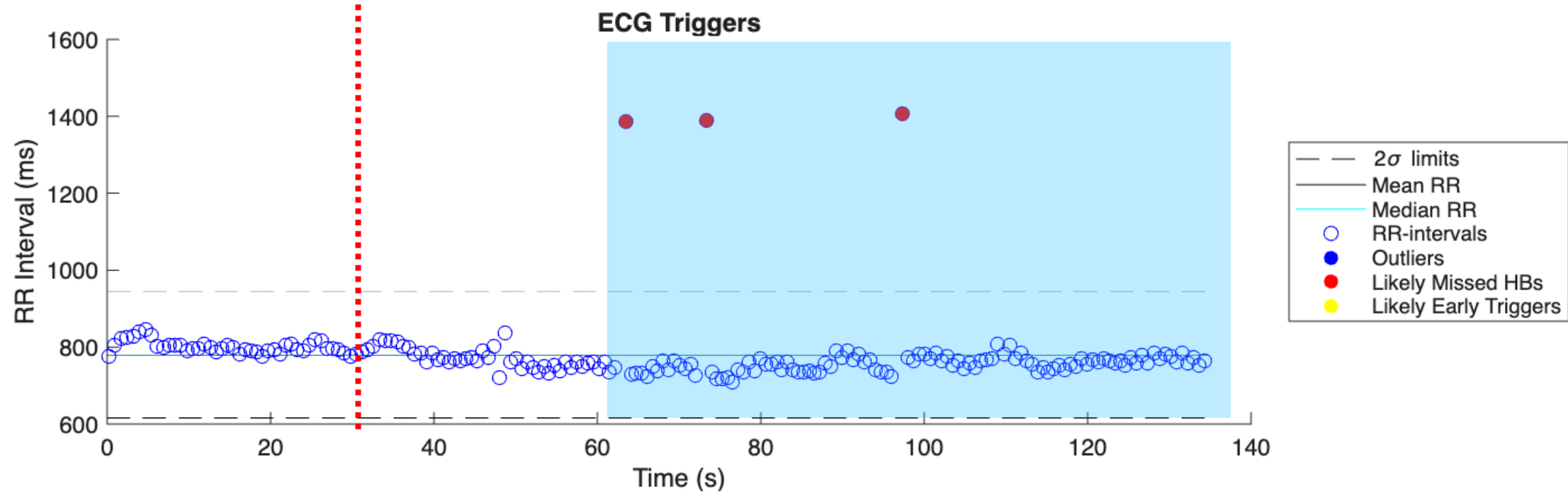
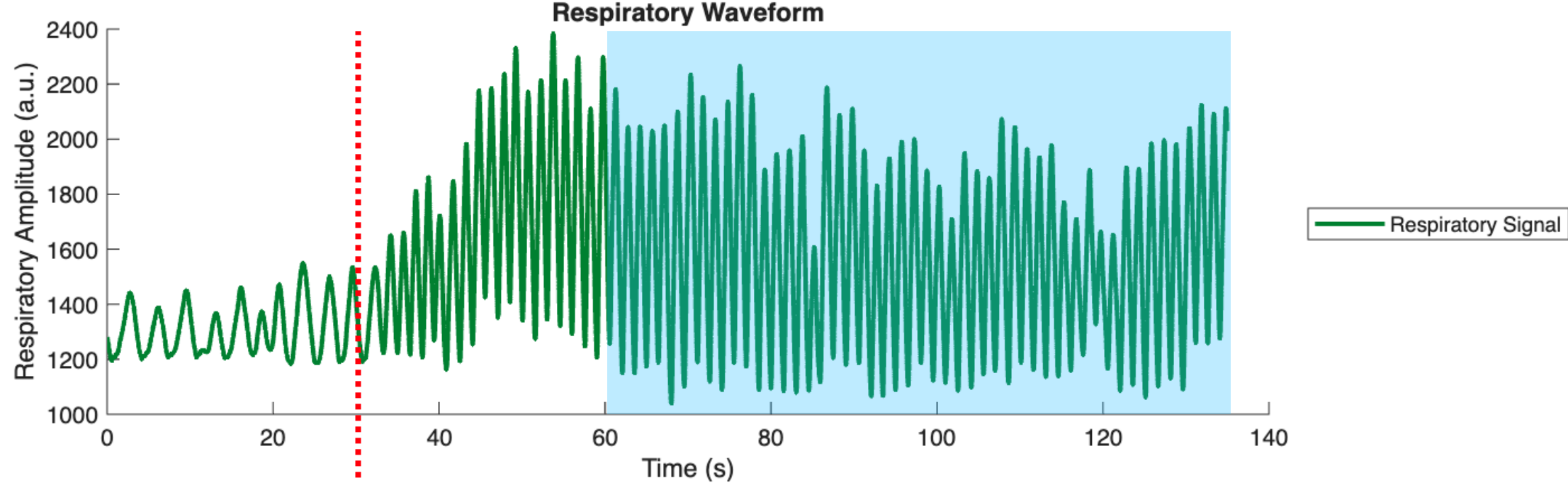


Radial



SMS





Free-breathing Exhalation

Free-breathing Inhalation

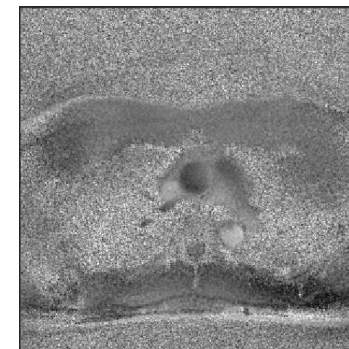
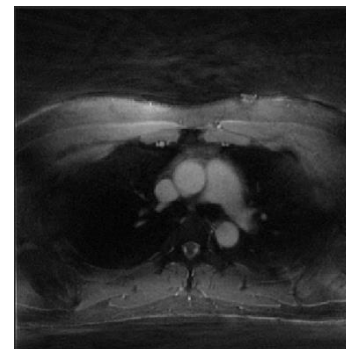
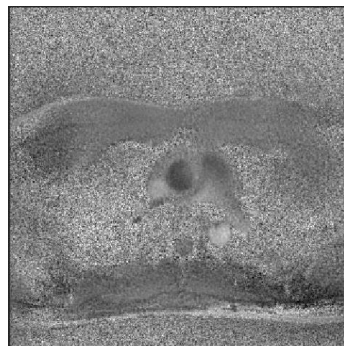
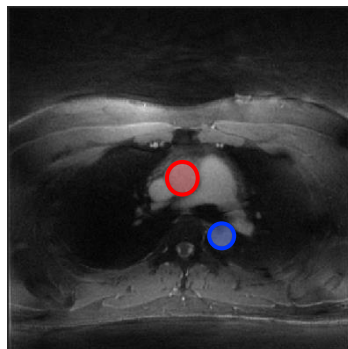
Magnitude

Phase

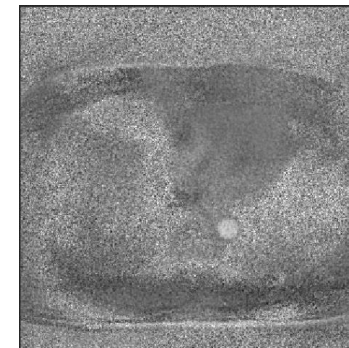
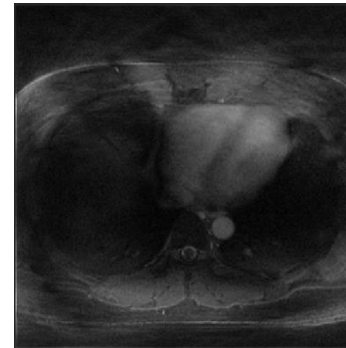
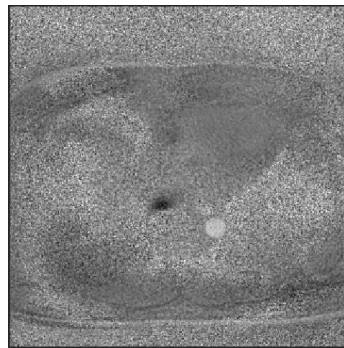
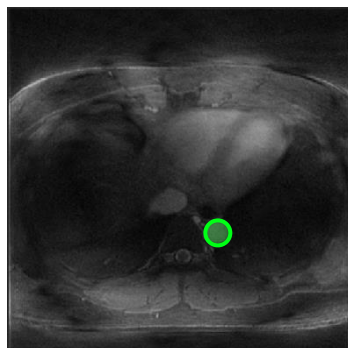
Magnitude

Phase

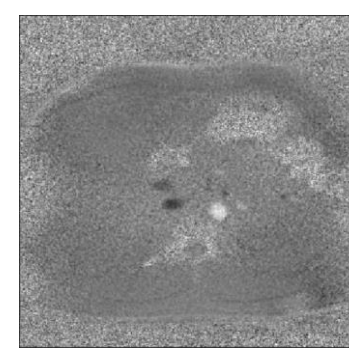
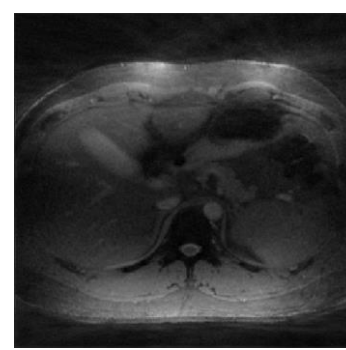
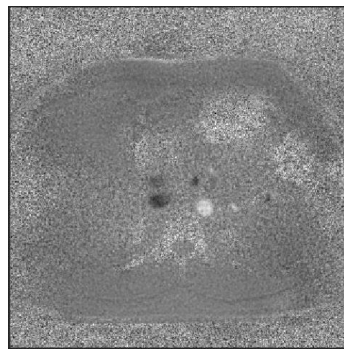
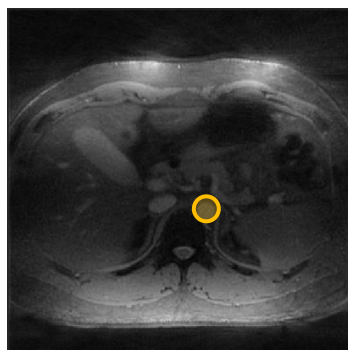
Aortic Arch



Thoracic Aorta



Abdominal Aorta

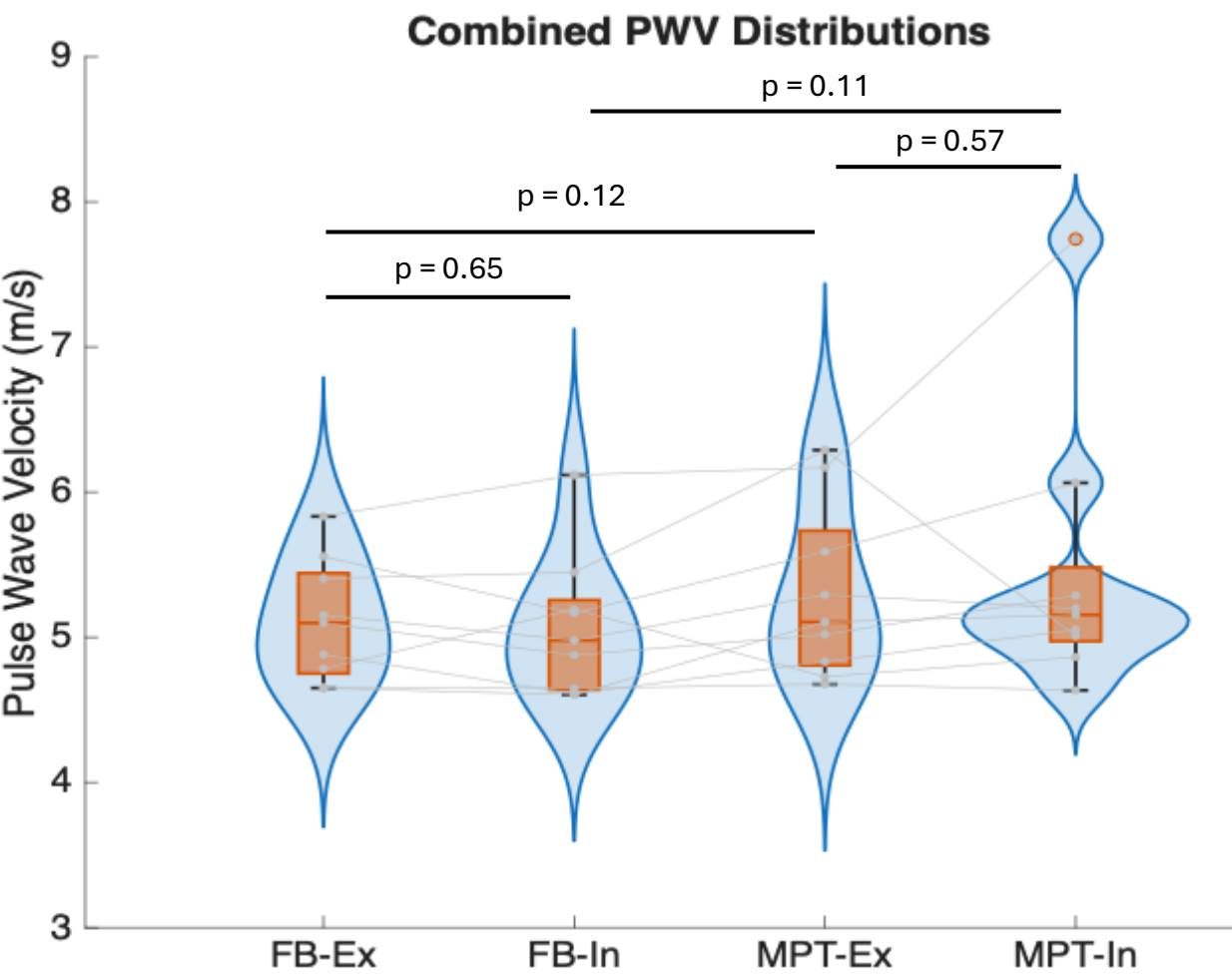


+150 cm/s



-150 cm/s

A



B

